TechNest – A Modular Front-End eCommerce Web Application

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Abstract

This project involved designing and building a front end eCommerce web application called TechNest. The application was created for a fictional electronics and gadgets retailer and runs entirely in the browser using HTML5, CSS3, Bootstrap 5 and JavaScript ES6+. There is no server or database involved at any point as all logic and data is handled on the client side.

The main problem this project aimed to address was the lack of structured, well organised code in typical student level front end projects. Many introductory web applications are written in a loose, procedural style with little thought given to code organisation or maintainability. TechNest was built to demonstrate that Object Oriented Programming principles and a modular file structure can be applied to a purely client side application in a way that reflects more professional development practices.

The finished application allows users to browse a product catalogue, filter and search for items, view individual product pages, manage a shopping basket and complete a checkout form. Basket data is saved using the browser's localStorage so it is not lost when moving between pages. The project was developed over a 10 week cycle with version control managed through Git and GitHub throughout.

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1. Introduction

1.1. Purpose

The purpose of this project is to design and develop a structured front-end eCommerce web application named TechNest. The application represents a fictional electronics and gadgets retailer and is implemented entirely within the browser using HTML5, CSS3, Bootstrap 5 and JavaScript (ES6+), without the use of any server-side backend or database.

TechNest enables users to browse products, filter and search a catalogue, view individual product detail pages, manage a shopping basket and complete a structured checkout process. All application logic, including basket management and data persistence, is handled client-side through modular JavaScript and the browser's localStorage API.

This report documents the complete development lifecycle of the project, including background research, system design, implementation decisions, testing strategy and overall evaluation.

1.2. Background

1.2.1. Scene Setting

Online retail has grown significantly over the past two decades and is now one of the primary methods by which consumers purchase goods. Statista (2024) reports that global eCommerce revenue reached approximately $4.1 trillion in 2024 and continues to increase annually. For smaller businesses in particular, establishing an online presence is no longer optional, as customers increasingly expect to browse and purchase products digitally.

Many commercial eCommerce platforms such as Shopify and WooCommerce rely heavily on server-side processing, database management and hosting infrastructure. While these platforms are powerful, they may be unnecessarily complex for small retailers who require a lightweight system primarily focused on product display and basket functionality. As modern browsers have become more capable, interest has grown in structured client-side solutions that can simulate realistic application behaviour without backend integration.

1.2.2. Research Background

JavaScript has evolved considerably since its early use as a simple scripting language. The introduction of ES6 brought class-based syntax, modules, arrow functions and template literals, making the language more suitable for structured and maintainable application development (MDN Web Docs, 2023).

Research into front-end architecture highlights the importance of separation of concerns, encapsulation and modular design. Osmani (2022) argues that applying design patterns and structured module organisation improves maintainability and testability. Similarly, Flanagan (2020) demonstrates that carefully organised client-side code can replicate complex application behaviour effectively, even without server-side components.

Responsive design principles are also central to modern web development. Since the introduction of responsive web design (Marcotte, 2010), frameworks such as Bootstrap have simplified cross-device interface development. Bootstrap 5 removed its dependency on jQuery and aligns with contemporary JavaScript development practices (Bootstrap, 2021).

1.2.3. Gap

Despite the availability of modern JavaScript features and responsive frameworks, many student-level web projects remain procedural in structure and lack architectural discipline. Limited attention is often given to modular file organisation, clear separation between data, business logic and presentation, or the use of Object-Oriented Programming principles in front-end systems.

This represents a gap in practical software engineering education, as structured code organisation is a core competency in professional development environments. By applying ES6 class-based OOP, modular JavaScript structure and client-side state management within a browser-only application, this project demonstrates that professional architectural practices are both achievable and beneficial in front-end development contexts.

1.3. Aims & Objectives

1.3.1. Aim

The aim of this project is to design and deliver a fully functional, modular front-end eCommerce web application that demonstrates the application of Object-Oriented Programming principles and structured client-side architecture within a defined 10-week development cycle.

1.3.2. Objectives

- To design and build a responsive multi-page web application using HTML5, CSS3, Bootstrap 5 and JavaScript ES6+.
- To implement ES6 class-based Object-Oriented Programming to encapsulate product data, basket logic and storage management.
- To separate business logic, data handling and user interface rendering into structured JavaScript modules.
- To implement dynamic product listing using DOM manipulation techniques.
- To add category filtering, keyword search and sorting functionality to the product catalogue.
- To implement full basket CRUD operations with automatic price recalculation.
- To use localStorage to persist basket data across page navigation.
- To create individual product detail pages using URL parameter handling.
- To implement a structured checkout form with client-side validation.
- To manage source code using Git with regular commits to a GitHub repository.
- To conduct functional testing against defined requirements and document results.

1.4. Report Outline

Chapter 2 – Design presents the functional and non-functional requirements of the application, user stories, wireframes and architectural decisions.

Chapter 3 – Implementation details the technical development of the system, including the ES6 classes, JavaScript modules and page structures used.

Chapter 4 – Testing outlines the testing methodology, test cases and recorded results.

Chapter 5 – Conclusion and Future Work evaluates the project outcomes, identifies limitations and proposes potential improvements.

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